# Chapter 11, Problem 48P

## Problem

Determine the complex power for the following cases:

(a) $P$ = 269 W, $Q$ = 150 VAR (capacitive)

(b) $Q$ = 2000 VAR, pf = 0.9 (leading)

(c) $S$ = 600 VA, $Q$ = 450 VAR (inductive)

(d) $V$rms = 220 V, $P$ = 1 kW,

$|\mathbf{Z}|$ = 40 Ω (inductive)

## Step-by-step solution

**Step 1** of 8

(a)

$$S = P - jQ$$

$$\boxed{S = 269 - j150\ \text{VA}}$$

**Step 2** of 8

(b)

$$\text{Pf} = \cos\theta$$

$$0.9 = \cos\theta$$

$$\theta = 25.84°$$

$$Q = 2000\ \text{VAR}$$

$$S = \frac{Q}{\sin\theta}$$

$$S = \frac{2000}{\sin(25.84°)}$$

$$S = 4588.31$$

**Step 3** of 8

$$P = S\cos\theta$$

$$P = 4588.31 \times \cos 25.84$$

$$P = 4129.48$$

Therefore,

$$S = P - jQ$$

$$\boxed{S = 4129.48 - j2000\ \text{VA}}$$

**Step 4** of 8

(c)

$S = 600\ \text{VA}$

$Q = 450\ \text{VAR}$

$S = \dfrac{Q}{\sin\theta}$

$\sin\theta = \dfrac{Q}{S}$

$\sin\theta = \dfrac{450}{600}$

$\sin\theta = 0.75$

$\theta = 48.59$

**Step 5** of 8

$P = S\cos\theta$

$P = 600\cos(48.59)$

$P = 396.86$

Therefore,

$\boxed{S = 396.86 + j450\ \ \text{VA}}$

**Step 6** of 8

(d)

$V_{\text{rms}} = 220\text{V},$

$P = 1\text{kW},$

$|Z| = 40\Omega$

$S = \dfrac{|V|^2}{|Z|}$

$S = \dfrac{(220)^2}{40}$

$S = 1210$

**Step 7** of 8

$P = S\cos\theta$

$\cos\theta = \dfrac{P}{S}$

$\cos\theta = \dfrac{1000}{1210}$

$\cos\theta = 0.826$

$\theta = 34.31$

**Step 8** of 8

$$Q = S\sin\theta$$

$$Q = 1210\sin(34.31)$$

$$Q = 682.041$$

Therefore,

$$\boxed{S = 1000 + j682.041 \quad \text{VA}}$$